Part 2b: Oligopoly Models

Instructor: Oleg Baranov

Microeconomics 2 *(Module 1)*

Suggested Readings for this part

Readings in Class Textbooks:

- **Varian, 8th edition:**
  - Ch 27: Oligopoly
- **Carlton and Perloff:**
  - Ch 6: Noncooperative Oligopoly
  - Ch 7: Product Differentiation

Models of oligopoly

- Two classical models with homogeneous goods
  - **Bertrand model — price-setting**: firms choose prices, a market demand determines a quantity.
  - **Cournot model — quantity-setting**: firms choose quantities, a market demand determines a price at which the market clears.
- The models lead to very different market predictions.
- This contrasts with a monopoly setting in which choosing **prices** or **quantities** is equivalent

Cournot Competition

Cournot competition: Setting

- N firms indexed by $i = 1, \ldots, N$
- The cost function of each firm is $C_i(q_i)$
- Firms simultaneously choose their quantities $q_1, q_2, ..., q_N$
- Total quantity is given by $Q = q_1 + q_2 + ... + q_N$
- The market price is given by an inverse demand function $p(Q)$
- Firm $i's$ profit is the following function of $q = (q_1, \ldots, q_N)$:

$$\pi_i(q) = p\left(\sum_{j \neq i} q_j + q_i\right) q_i - C_i(q_i).$$

Cournot competition: Solution

- We look for a **pure strategy Nash equilibrium**. **How to solve for a Nash equilibrium here?**

1. Fix arbitrary q_{-i} and solve for the firm i's best response, i.e.

$$\max_{q_i \geq 0} \ \pi_i(q_i, q_{-i}).$$

   Again: **FOC, SOC, shutdown** $\implies BR_i(q_{-i})$.

2. Then solve the system of equations in N variables $q^* = (q_1^*, \ldots, q_N^*)$:

$$q_i^* \ = \ BR_i(q_{-i}^*) \qquad i \ = \ 1, \ldots, N \ .$$

Class Example 1

- Inverse market demand is given by

$$p(Q) = 8 - Q$$

- N firms, each with $C(q) = 2q$
- Firms compete in **quantities**

- **Question 1:** Solve for a market outcome when $N = 1$.
- **Question 2:** Solve for a market outcome when $N = 2$.
- **Question 3:** For $N = 2$, draw the best response diagram in the (q_1, q_2) coordinates.

Cournot competition: Intuition

- The optimal production level Q^m for the monopoly is given by

$$MR(Q^*) \ = \ p\left(Q^m\right) \ + \ p'\left(Q^m\right) Q^m \ = \ MC(Q^m)$$

- In the Nash equilibrium, for any firm i with $q_i^* > 0$, it must be

$$p\left(Q^*\right) \ + \ p'\left(Q^*\right) q_i^* \ = \ MC_i(q_i^*)$$

  where

$$Q^* \ = \ q_1^* \ + \ q_2^* \ + \ \dots \ + \ q_N^*$$

  Intuition 1: "*marginal revenue equals marginal costs*" but only accounting for your own share of the output.

- **Intuition 2:** a Cournot firm acts as a monopoly on the residual demand

$$\tilde{p}_i(q) \ = \ p(Q_{-i}^* \ + \ q)$$

Cournot competition: Final Remarks

- Some firms with high costs might shutdown voluntarily in the equilibrium.
- Typically, with nondecreasing marginal costs and no fixed costs, a NE is unique.
- With fixed costs, multiple equilibria are possible including the ones with different number of active firms (firm is active when $q_i^* > 0$).
- Firms are better off compared to perfect competition, but worse off compared to the monopoly setting

Bertrand Competition

Bertrand competition: Setting

- Two firms indexed by $i = 1, 2$, with cost function $C_i(q_i)$
- The market demand function is $D(p)$
- Firms simultaneously choose prices, p_1 and p_2. Let's assume that:
  - Consumers purchase from the firm with the **lowest price**.
  - When price is the same, they **split the market equally**.
- Firm i's profit as a function of p_i, p_j:

$$\pi_i(p_i, p_j) = \begin{cases} p_i \cdot D(p_i) - C_i(D(p_i)) & p_i < p_j \\ p_i \cdot D(p_i)/2 - C_i(D(p_i)/2) & p_i = p_j \\ -C_i(0) & p_i > p_j \end{cases}$$

Bertrand competition: Solution

- We look for a pure strategy Nash equilibrium $p^* = (p_1^*, p_2^*)$.
- We can work with the best responses as before, that is

$$\max_{p_i \geq 0} \ \pi_i(p_i, p_j) \qquad \Longrightarrow \qquad BR_i(p_j).$$

- **But the profit function is non-continuous and best response is undefined for certain ranges**
- In Bertrand models, it is easier to find NE by ruling out non-equilibria with educated observations
  - o Firms with positive sales have to charge the same price;
  - o Firms should not lose money when they sell ($\pi_i^* \geq -C_i(0)$)
  - o No incentives for undercutting *(charging a slightly lower price)*

Class Example 2

- Market demand is given by

$$D(p) = 8 - p$$

- Two firms, each with $C(q) = 2q$
- Firms compete in **prices**

- **Question 1:** Solve for a market outcome.

Remarks on Bertrand

- With constant marginal costs, the NE is either unique, or does not even exist
- NE is usually not unique when marginal costs are increasing *(due to weaker incentives to engage in undercutting)*
- All firms with high costs shutdown — costs advantage is **crucial** in Bertrand setting.
- **Bertrand paradox:** Set of equilibria frequently includes (or even consists of) the one corresponding to the perfect competition outcome

Bertrand paradox

Bertrand paradox: with homogeneous good and constant marginal costs, Bertrand equilibrium implements a perfectly competitive outcome.

In practice, it does not play out because:

- Discrete prices
- Dynamic interaction & collusion
- Consumers' search *(can be hard to find the best price)*
- firms' capacity constraints *(can be hard to serve all customers)*
- **Product differentiation:** vertical *(different quality)* or horizontal *(different features, same quality)*

Hotelling Competition

Hotelling Model ("Hotelling Beach")

- A beach town with a **1km broadwalk**
- Two identical firms located at $0 \leq a_1 \leq a_2 \leq 1$
- Each firm can produce the good at constant marginal cost of c
- Consumers with unit demand (each with **wtp** of w)
- Consumers are evenly spread out along the broadwalk (uniform distribution)
- Consumers can buy from either firm, but they have to walk first. Walking costs $t \cdot d$ for any distance d (in monetary equivalent)
- Let's assume

$$t > 0 \quad \text{and} \quad w > c + t$$

What would a monopoly do here?

- Let's assume that it is optimal for a monopoly to serve all customers (w is large enough).
- Where should the monopoly place its shop?
- Fix location at $a \in [0,1]$. The longest travel for a consumer is given by $\max\{a, 1-a\}$. To serve everyone, the optimal monopoly price is given by

$$p = w - t \cdot \max\{a, 1-a\}$$

- The best location when serving everyone is $a = 0.5$
- In the absence of competition, firms go to consumers to minimize their costs (and to maximize the surplus to be extracted).

  This is the principle of minimum differentiation

Hotelling Model: Same Location

- Two firms located at the same location ($a_1 = a_2$)
- Then we have the same product, the same location and firms competing in prices
- **This is the Bertrand model**
- The unique Nash equilibrium:

$$p_1^* = p_2^* = c$$

and

$$\pi_1^* = \pi_2^* = 0$$

Back to Hotelling Model

- Two firms are located at the ends of the broadwalk ($a_1 = 0$, $a_2 = 1$)
- Let's assume that all consumers will buy the good (w is large enough, need to check later)
- Solving for an indifferent consumer *(same value of buying)*

$$\begin{aligned} w - p_1 - t\hat{x} &= w - p_2 - t(1-\hat{x}) \\ \hat{x} &= \tfrac{1}{2} + \tfrac{p_2 - p_1}{2t} \end{aligned}$$

- Then demands for each firm are given by

$$D_1(p_1, p_2) = \hat{x} = \tfrac{1}{2} + \tfrac{p_2 - p_1}{2t} \qquad D_2(p_1, p_2) = 1 - \hat{x} = \tfrac{1}{2} + \tfrac{p_1 - p_2}{2t}$$

- And profits of firms are given by

$$\pi_1(p_1, p_2) = (p_1 - c) \cdot D_1(p_1, p_2) \qquad \pi_2(p_1, p_2) = (p_2 - c) \cdot D_2(p_1, p_2)$$

Hotelling Model: Solution

- We look for a pure strategy Nash equilibrium $p^* = (p_1^*, p_2^*)$.
- Fix p_j and solve for firm i's best response, i.e.,

$$\max_{p_i \geq 0} \ \pi_i(p_i, p_j) \qquad \Longrightarrow \qquad BR_i(p_j).$$

- Solve for (p_1^*, p_2^*) such that

$$p_1^* \ = \ BR_1(p_2^*) \qquad \text{and} \qquad p_2^* \ = \ BR_2(p_1^*)$$

Hotelling Model: Nash equilibrium

- Best response functions are given by

$$BR_1(p_2) = \frac{t + c + p_2}{2} \quad \text{and} \quad BR_2(p_1) = \frac{t + c + p_1}{2}$$

- Then Nash equilibrium (p_1^*, p_2^*)

$$p_1^* = p_2^* = t + c$$

- Equilibrium profits are

$$\pi_1^* = \pi_2^* = \frac{t}{2} > 0$$

- **Last check:** max cost in NE is $p^* + t/2$, so we need

$$w \geq c + \frac{3}{2}t$$

Discussion

- In the Hotelling model, prices and profits increase with
  - the distance between firms; and
  - the cost of traveling t.
- When facing competition, firms differentiate themselves as much as possible

  The principle of maximum differentiation
- We do not have to think about consumer location as literally *"physical location"*. Instead, we can think of $x \in [0, 1]$ as a bundle of characteristics, i.e., color, taste, shape, convenience, inconvenience, ambience.

Differentiated Products Model

Alternative Model of Differentiated Products

- Two firms $i = 1, 2$, each with cost function $C_i(q_i)$
- Market demand for Firm i depends on p_i and p_j

$$D_i(p_i,\ p_j) \ = \ 1 \ - \ bp_i \ + \ ap_j$$

where

- o $b > 0$ — negative impact of p_i on D_i (as usual)
- o $a > 0$ — goods are substitutes $(p_j \uparrow \ \Rightarrow \ D_i \uparrow)$
- o $a < 0$ — goods are complements $(p_j \uparrow \ \Rightarrow \ D_i \downarrow)$
- o $a = 0$ — unrelated goods

- Firm i's profit is given by

$$\pi_i(p_i, p_j) \ = \ p_i \cdot D_i(p_i, p_j) \ - \ C_i(D_i(p_i, p_j))$$

- You can find the NE using the standard solution process *(the one we used for solving Cournot and Hotelling models)*

Skill checklist for Part 2b

1. Solving for NE in various Cournot models
2. Solving for NE in various Bertrand models
3. Solving for NE in various Hotelling models
4. Solving for NE in the Differentiated Products model
5. Calculating various market characteristics for each model